Cellular Respiration

The Process of Energy Production in Cells

Cellular respiration is the process by which cells convert glucose and oxygen into energy (ATP), carbon dioxide, and water. This fundamental process occurs in all living organisms and is essential for life.

The overall equation for cellular respiration is:

$C_6H_{12}O_6 + 6O_2 \cdot 6CO_2 + 6H_2O + ATP$ (energy)

This process occurs in three main stages:
1. Glycolysis (in the cytoplasm)
2. Krebs Cycle (in the mitochondrial matrix)
3. Electron Transport Chain (in the inner mitochondrial membrane)

Glycolysis: The First Stage

Cell

Cytoplasm

Glucose · Pyruvate

Glycolysis is the first stage of cellular respiration. It occurs in the cytoplasm of the cell and does not require oxygen (anaerobic process).

Key points about glycolysis:
- glucose (6 carbons) is broken down into 2 pyruvate molecules (3 carbon
- Net production: 2 ATP and 2 NADH
- Occurs in the cytoplasm
- Does not require oxygen

The process involves 10 enzymatic reactions, which can be divided into tw

1. Energy Investment Phase (steps 1-5): Uses 2 ATP
2. Energy Payoff Phase (steps 6-10): Produces 4 ATP and 2 NADH